

torch.histogramdd#

<https://pytorch.org/docs/stable/generated/torch.histogramdd.html#torch.histogramdd>

Description:

Computes a multi-dimensional histogram of the values in a tensor. Interprets the elements of an input tensor whose innermost dimension has size N as a collection of N -dimensional points. Maps each of the points into a set of N -dimensional bins and returns the number of points (or total weight) in each bin. input must be a tensor with at least 2 dimensions. If input has shape (M, N) , each of its M rows defines a point in N -dimensional space. If input has three or more dimensions, all but the last dimension are flattened. Each dimension is independently associated with its own strictly increasing sequence of bin edges. Bin edges may be specified explicitly by passing a sequence of 1D tensors. Alternatively, bin edges may be constructed automatically by passing a sequence of integers specifying the number of equal-width bins in each dimension. corresponding to its dimension

Function Signature

```
torch.histogramdd(input, bins, *, range=None, weight=None, density=False, out=None) -> (Tensor, Tensor[])#
```

For each N-dimensional point in input:

Each of its coordinates is binned independently among the bin edges

Binning results are combined to identify the N-dimensional bin (if any)

Parameters

Parameters

Keyword

range (sequence of float) – Defines the leftmost and rightmost bin edges in each dimension. weight (Tensor) – By default, each value in the input has weight 1. If a weight tensor is passed, each N-dimensional coordinate in input contributes its associated weight towards its bin's result. The weight tensor should have the same shape as the input tensor excluding its innermost dimension N. density (bool) – If False (default), the result will contain the count (or total weight) in each bin. If True, each count (weight) is divided by the total count (total weight), then divided by the volume of its associated bin.

Returns:

hist (Tensor)

Code Examples

```
>>> torch.histogramdd(torch.tensor([[0., 1.], [1., 0.], [2., 0.], [2., 2.]]), bins=[3, 3],
...                    weight=torch.tensor([1., 2., 4., 8.]))
torch.return_types.histogramdd(
  hist=tensor([[0., 1., 0.],
               [2., 0., 0.],
               [4., 0., 8.]]),
  bin_edges=(tensor([0.0000, 0.6667, 1.3333, 2.0000]),
             tensor([0.0000, 0.6667, 1.3333, 2.0000])))

>>> torch.histogramdd(torch.tensor([[0., 0.], [1., 1.], [2., 2.]]), bins=[2, 2],
...                    range=[0., 1., 0., 1.], density=True)
torch.return_types.histogramdd(
  hist=tensor([[2., 0.],
               [0., 2.]]),
  bin_edges=(tensor([0.0000, 0.5000, 1.0000]),
             tensor([0.0000, 0.5000, 1.0000])))
```

Notes & Warnings

NOTE

Note See also `torch.histogram()`, which specifically computes 1D histograms. While `torch.histogramdd()` infers the dimensionality of its bins and binned values from the shape of input, `torch.histogram()` accepts and flattens input of any shape.

Detailed Documentation

Documentation

Computes a multi-dimensional histogram of the values in a tensor.

Interprets the elements of an input tensor whose innermost dimension has size N as a collection of N -dimensional points. Maps each of the points into a set of N -dimensional bins and returns the number of points (or total weight) in each bin.

input must be a tensor with at least 2 dimensions. If input has shape (M, N) , each of its M rows defines a point in N -dimensional space. If input has three or more dimensions, all but the last dimension are flattened.

Each dimension is independently associated with its own strictly increasing sequence of bin edges. Bin edges may be specified explicitly by passing a sequence of 1D tensors. Alternatively, bin edges may be constructed automatically by passing a sequence of integers specifying the number of equal-width bins in each dimension.

corresponding to its dimension

into which the point falls

If the point falls into a bin, the bin's count (or total weight) is incremented

Points which do not fall into any bin do not contribute to the output

bins can be a sequence of N 1D tensors, a sequence of N ints, or a single int.

If bins is a sequence of N 1D tensors, it explicitly specifies the N sequences of bin edges. Each 1D tensor should contain a strictly increasing sequence with at least one element. A sequence of K bin edges defines K-1 bins, explicitly specifying the left and right edges of all bins. Every bin is inclusive of its left edge. Only the rightmost bin is inclusive of its right edge.

If bins is a sequence of N ints, it specifies the number of equal-width bins in each dimension. By default, the leftmost and rightmost bin edges in each dimension are determined by the minimum and maximum elements of the input tensor in the corresponding dimension. The range argument can be provided to manually specify the leftmost and rightmost bin edges in each dimension.

If bins is an int, it specifies the number of equal-width bins for all dimensions.

See also `torch.histogram()`, which specifically computes 1D histograms. While `torch.histogramdd()` infers the dimensionality of its bins and binned values from the shape of input, `torch.histogram()` accepts and flattens input of any shape.

input (Tensor) – the input tensor.

bins – Tensor[], int[], or int. If Tensor[], defines the sequences of bin edges. If int[], defines the number of equal-width bins in each dimension. If int, defines the number of equal-width bins for all dimensions.

range (sequence of float) – Defines the leftmost and rightmost bin edges in each dimension.

weight (Tensor) – By default, each value in the input has weight 1. If a weight tensor is passed, each N-dimensional coordinate in input contributes its associated weight towards its bin's result. The weight tensor should have the same shape as the input tensor excluding its innermost dimension N.

density (bool) – If False (default), the result will contain the count (or total weight) in each bin. If True, each count (weight) is divided by the total count (total weight), then divided by the volume of its associated bin.

N-dimensional Tensor containing the values of the histogram. bin_edges(Tensor[]): sequence of N 1D Tensors containing the bin edges.